

- B. Check the column shear capacity assuming failure along two shear planes inclined at 45-degree angles above and below the point of force application.

Commentary: Based on the observed failure mechanism of pier columns involved in large truck collisions, an acceptable method of calculating the column strength to resist the vehicular collision force is to assume failure along two shear planes inclined at 45-degree angles above and below the point of force application. Reference Research Report No. FHWA/TX-10/9-4973-1.

2.7 FORCE EFFECTS DUE TO SUPERIMPOSED DEFORMATIONS (LRFD 3.12)

2.7.1 Uniform Temperature

- A. In lieu of **LRFD** 3.12.2, Procedures A and B, substitute the following table:

Table 2.7.1-1 Temperature Range by Superstructure Material

Superstructure Material	Temperature Range (Degrees Fahrenheit)			
	Mean	High	Low	Range
Concrete Only	70	105	35	70
Concrete Deck on Steel Girder	70	110	30	80
Steel Only	70	120	30	90

- B. Note the minimum and maximum design temperatures on drawings for girders, expansion joints and bearings.
- C. For detailing purposes, take the normal mean temperature from this table.
- D. In accordance with **LRFD** Table 3.4.1-1, base temperature rise and fall on 120% of the maximum value given in Table.

2.7.2 Temperature Gradient (LRFD 3.12.3)

Delete the second paragraph of **LRFD** 3.12.3 and substitute the following:

"Include the effects of Temperature Gradient in the design of continuous concrete superstructures only. The vertical Temperature Gradient shall be taken as shown in **LRFD** Figure 3.12.3-2."

2.8 BARRIERS AND RAILINGS (LRFD 4.6.2.2)

2.8.1 Distribution for Beam-Slab Bridges

Distribute barrier and railing permanent loads in accordance with **LRFD** 4.6.2.2.

*Commentary: This **LRFD** requirement is repeated here for clarity because previous versions of the **SDG** contained a different methodology for distributing railing loads across the width of the bridge deck.*